

		acceleration = 0.212 m s^{-2}			
				Total	
					B1 [3]
4	(a)	(i) $(p =) mv$			B1
		(ii) $E_k = \frac{1}{2}mv^2$			B1
		algebra leading to			M1
		$E_k = p^2/2m$			A0 [3]
	(b)	(i) $\Delta p = 0.035 (4.5 + 3.5)$ OR $a = (4.5 + 3.5)/0.14$			C1
		$= 0.28 \text{ N s}$ $= 57.1 \text{ m s}^{-2}$			
		force = $\Delta p / \Delta t (= 0.28/0.14)$ OR $F = ma (= 0.035 \times 575.1)$ (allow e.c.f.)			C1
		$= 2.0 \text{ N}$			A1
		<i>Note: candidate may add $mg = 0.34 \text{ N}$ to this answer, deduct 1 mark upwards</i>			B1 [4]
		(ii) $\text{loss} = \frac{1}{2} \times 0.035 (4.5^2 - 3.5^2)$			C1
		$= 0.14 \text{ J}$			A1 [2]
		(No credit for $0.28^2/(2 \times 0.035) = 1.12 \text{ J}$)			
	(c)	e.g. plate (and Earth) gain momentum			
		<i>i.e. discusses a 'system'</i>			B1
		equal and opposite to the change for the ball			
		<i>i.e. discusses force/momentum</i>			M1
		so momentum is conserved			
		<i>i.e. discusses consequence</i>			A1 [3]
				Total	[12]